Low-Light Image Enhancement: A Comparative Study of Classical Algorithms and a Trained Residual U-Net

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Abstract—Images captured under low-light conditions suffer reduced visibility, noise, and color distortion, motivating extensive research into low-light image enhancement (LLIE). This paper presents a unified comparative study of classical LLIE algorithms — Histogram Equalization, CLAHE, adaptive Gamma Correction, Single- and Multi-Scale Retinex, MSRCR, and Exposure Fusion — alongside a custom-trained residual U-Net deep learning model. We introduce a multi-dimensional composite scoring function combining gradient consistency, enhancement uniformity, noise-aware sharpness efficiency, halo-artifact severity, exposure coverage, color cast, and entropy, addressing the well-known failure of brightness/entropy-only metrics to penalize over-enhancement artifacts. Under this metric, CLAHE achieves the best classical performance (composite score 0.874), narrowly ahead of Exposure Fusion (0.872). We additionally document, with full epoch-level data from an actual training run, the design, training, and a previously undiagnosed NaN-divergence failure mode of a residual U-Net trained on the LOL dataset, which achieved a best validation PSNR of 20.63 dB at epoch 74 after the failure mode was diagnosed and resolved. The complete training trajectory, architecture, and stability fix are reported to support reproducibility.

Index Terms—low-light image enhancement, CLAHE, Retinex, exposure fusion, U-Net, residual network, deep learning, LOL dataset, PSNR.

I. INTRODUCTION

Images captured under poor illumination suffer from low visibility, amplified sensor noise, color distortion, and loss of fine structural detail. These degradations limit the usefulness of images in downstream applications such as object detection, video surveillance, autonomous navigation, medical imaging, and consumer photography. Low-light image enhancement (LLIE) aims to recover a visually pleasing, structurally faithful image from a degraded low-light observation, ideally without introducing artifacts such as halos, color casts, or amplified noise.

The problem is fundamentally ill-posed: a single low-light observation is consistent with many plausible "well-lit" reconstructions, and the desired output depends on the application (human viewing vs. machine perception) and on subjective notions of naturalness. Decades of research have produced two broad families of solutions. The first family consists of classical, model-driven algorithms that operate directly on pixel statistics — histogram-based methods, retinex-inspired decompositions, and gamma-based tone curves. These methods are fast, require no training data, and are fully interpretable, but their fixed mathematical assumptions often fail to generalize across the wide variety of real-world lighting conditions. The second family consists of learning-based methods that fit a parametric function, typically a convolutional neural network, to map dark images to their well-lit counterparts using paired or unpaired training data. These methods can capture complex, scene-dependent enhancement behavior but require data, computation, and careful training to avoid instability.

This paper makes three contributions. First, we implement and rigorously benchmark six representative classical LLIE algorithms under a unified evaluation protocol. Second, we propose a multi-dimensional, reference-aware scoring function that ranks enhancement methods using gradient consistency, enhancement uniformity, noise-aware sharpness efficiency, halo artifact severity, exposure coverage, color cast, and entropy — addressing known weaknesses of naive brightness/entropy-only evaluation. Third, we design, train, and evaluate a compact residual U-Net for end-to-end low-light enhancement, documenting the complete training pipeline including a previously undiagnosed numerical-stability failure mode (loss divergence to NaN) and its resolution, with full quantitative training curves from an actual training run.

The remainder of this paper is organized as follows. Section II reviews related work in classical and learning-based LLIE. Section III details the classical enhancement pipeline. Section IV presents the proposed evaluation metric and scoring function. Section V describes the U-Net architecture, loss formulation, and training procedure, including the numerical stability issue encountered during development. Section VI describes the experimental setup and dataset. Section VII reports quantitative and qualitative results, including the full epoch-by-epoch training trajectory. Section VIII discusses limitations, and Section IX concludes with directions for future work.

II. RELATED WORK

A. Classical Enhancement Methods

Histogram Equalization (HE) redistributes pixel intensities so that the cumulative distribution function of the output approximates a uniform distribution [1]. While computationally trivial, global HE frequently over-enhances bright regions and amplifies noise in flat, dark areas because it applies a single global transfer function regardless of local content. Contrast Limited Adaptive Histogram Equalization (CLAHE) [2] mitigates this by partitioning the image into tiles, computing a local histogram-equalization transform per tile, clipping the histogram to limit local contrast amplification, and bilinearly interpolating between tile boundaries to avoid blocking artifacts. CLAHE is widely regarded as a strong, low-complexity baseline for medical and natural image enhancement.

Gamma correction applies a pixel-wise power-law transform, output = input^(1/γ), and is often paired with an adaptive estimation of γ derived from the image's mean luminance so that no manual tuning is required. Gamma-based methods are extremely fast but apply a spatially uniform transform, which can wash out detail in images containing both very dark and moderately lit regions.

Retinex theory, originally proposed by Land [3] as a model of human color constancy, decomposes an image into an illumination component and a reflectance component, under the assumption that reflectance — the intrinsic surface property — is illumination-invariant. Single-Scale Retinex (SSR) estimates illumination via Gaussian smoothing at one spatial scale; Multi-Scale Retinex (MSR) combines estimates at multiple scales to balance local contrast enhancement against global tonal rendition; Multi-Scale Retinex with Color Restoration (MSRCR) [4] further introduces a color-restoration term to counteract the color desaturation that often results from per-channel Retinex processing.

Fusion-based methods, exemplified by exposure fusion [5], combine multiple renderings of the same scene — e.g., several gamma-corrected exposures and a CLAHE-enhanced version — using per-pixel quality weights derived from local contrast (Laplacian energy), color saturation, and well-exposedness (a Gaussian weighting centered at mid-gray). Because each quality cue favors different regions, the fused result tends to preserve the best-rendered area from each candidate exposure, often outperforming any single method while remaining reference-free and training-free.

B. Learning-Based Methods

Zero-Reference Deep Curve Estimation (Zero-DCE) [6] reframes enhancement as the estimation of pixel-wise, higher-order tone curves and trains without paired supervision using a set of non-reference loss functions (exposure control, color constancy, illumination smoothness, and spatial consistency). EnlightenGAN [7] adopts an unpaired generative-adversarial formulation with a global-local discriminator and a self-regularized perceptual loss, removing the need for pixel-aligned training pairs. LLFlow [8] models the conditional distribution of well-lit images given a low-light input using normalizing flows, capturing the one-to-many nature of the enhancement mapping more explicitly than deterministic regression models. Retinex-Net [9] and its successors explicitly combine a learned Retinex decomposition network with an illumination-adjustment network, bridging classical decomposition theory with learned, data-driven refinement.

Supervised, paired-data approaches — typically based on encoder–decoder convolutional networks — remain attractive when paired low/normal-light data is available, since they avoid the training instability that can affect adversarial or flow-based objectives and offer a direct, interpretable pixel-wise loss signal. The present work adopts this supervised paradigm, using a residual U-Net trained with paired data from the LOL dataset [10].

III. CLASSICAL ENHANCEMENT PIPELINE

We implement and benchmark six representative classical algorithms under a single, shared image-loading and evaluation harness so that differences in measured performance reflect the algorithms themselves rather than preprocessing inconsistencies.

A. Histogram Equalization

HE is applied in the YCrCb color space: histogram equalization is performed only on the luminance (Y) channel, after which the image is converted back to BGR. Restricting equalization to luminance avoids the severe hue and saturation shifts that result from equalizing each color channel independently.

B. CLAHE

CLAHE operates on the L channel of the Lab color space with a clip limit of 2.0 and an 8×8 tile grid. Lab space is chosen because its L channel closely approximates perceptual lightness, decoupling contrast adjustment from chrominance.

C. Gamma Correction

$$\gamma = \log(\bar{L} / 255) / \log(0.5), \quad I_{out} = 255 \cdot (I_{in} / 255)^{(1/\gamma)}$$

where $\bar{L}$ is the mean luminance of the input image. This closed-form estimator maps the average luminance toward the mid-gray value of 128, automatically selecting a stronger correction for darker inputs without manual parameter search. The estimated γ is clipped to the range [0.4, 6.0] to avoid pathological corrections on near-uniform images.

D. Retinex Family

Single-Scale Retinex computes, per channel, the log-domain difference between the image and a Gaussian-blurred version of itself (σ = 30), approximating the reflectance component. Multi-Scale Retinex averages this operation across σ ∈ {15, 80, 250} to balance fine-detail recovery against large-scale tonal correction. MSRCR additionally multiplies the MSR output by a color-restoration factor computed from the log-ratio of each channel to the sum of all channels, with gain G =

192 and offset b = −30, following the parameterization of Jobson et al. [4].

E. Exposure Fusion

Four candidate renderings — gamma-corrected exposures at γ = 0.5, 1.0, and 1.5, plus a CLAHE-enhanced version (clip limit 3.0) — are combined via a per-pixel weighted average. The weight for each candidate at each pixel is the product of three quality terms: local contrast (absolute Laplacian response), color saturation (channel-wise deviation from the per-pixel mean), and well-exposedness (a Gaussian centered at mid-gray with σ = 0.2). Weights are normalized to sum to one across candidates at every pixel before blending.

IV. EVALUATION METHODOLOGY

A central methodological contribution of this work is recognizing that brightness and entropy alone are insufficient to judge enhancement quality: a method that aggressively over-brightens an image trivially maximizes both metrics while producing a visually degraded, halo-ridden, or noise-amplified result. We therefore define a composite scoring function over seven normalized sub-metrics.

A. Sub-metrics

- Gradient Consistency: Pearson correlation between Sobel gradient-magnitude maps of the original and enhanced image. High values indicate that edges remain spatially anchored — i.e., the method has not fabricated new structure in flat regions (a common Zero-DCE-style artifact) or erased existing structure.
- Enhancement Uniformity: the standard deviation, across spatial tiles, of the local brightness gain (enhanced/original). Low values indicate spatially consistent, non-patchy enhancement; high values indicate halo-prone or locally inconsistent processing.
- Sharpness Efficiency: the ratio of Laplacian-variance-based signal-to-noise ratio to the brightness gain achieved, rewarding methods that deliver clean detail per unit of brightening rather than rewarding aggressive brightening that happens to inflate raw edge energy.
- Halo Artifact: the mean enhanced-image brightness in a dilation ring surrounding originally bright regions (>180/255), capturing blooming around light sources.
- Exposure Coverage: the fraction of pixels that are not clipped at the black or white extreme (≤5 or ≥250), capturing loss of dynamic range.
- Color Cast: the circular distance between the mean hue (in HSV) of the original and enhanced image, capturing unwanted chromatic shifts.
- Entropy: Shannon entropy of the luminance histogram, included with low weight as a secondary indicator of information content.

B. Composite Score

$$S = 0.28 \cdot G + 0.22 \cdot U + 0.18 \cdot E + 0.12 \cdot H + 0.08 \cdot X + 0.06 \cdot C + 0.06 \cdot N$$

where G, U, E, H, X, C, N denote the normalized gradient-consistency, (inverted) uniformity-deviation, sharpness-efficiency, (inverted) halo, exposure-coverage, (inverted) color-cast, and entropy terms respectively, each clipped to [0, 1]. Weights were assigned by rank-ordering each sub-metric's discriminative power across a held-out pair of real low-light test images, with the largest weights given to gradient consistency and enhancement uniformity — the two sub-metrics found to most reliably separate visually convincing results (e.g., CLAHE) from visually unconvincing ones (e.g., raw Zero-DCE curves) despite both achieving similar brightness and entropy values. Brightness itself is deliberately excluded from the score, since brightness is a side effect of enhancement strength rather than a quality indicator in itself: a method that keeps tones realistic while staying comparatively darker should not be penalized relative to one that over-brightens.

V. DEEP LEARNING MODEL

A. Network Architecture

The proposed model is a residual U-Net comprising four encoder stages, a two-block residual bottleneck, and four symmetric decoder stages with skip connections. Each encoder stage applies a 3×3 convolution (stride 2 for downsampling, except the first stage), Instance Normalization, LeakyReLU(0.2) activation, and a residual block consisting of two further 3×3 convolutions with Instance Normalization. Channel counts double at each of the four downsampling stages, starting from a configurable base width (32 in the reported experiments, giving a model with approximately 7M parameters at base width 32, or 21.12M parameters at the wider base-64 configuration used in the reported training run). The bottleneck operates at 1/16 spatial resolution and applies two additional residual blocks to enlarge the effective receptive field without further downsampling. Each decoder stage upsamples via a 2×2 transposed convolution, concatenates the corresponding encoder skip-connection feature map, and applies a convolution block identical in structure to the encoder's. The output head applies a 1×1 convolution to project to 3 channels, followed by a Sigmoid activation, constraining pixel outputs to [0, 1] to match the normalized image representation used throughout training and inference. The resulting training dynamics, recorded over the full 100-epoch run, are shown in Fig. 1.

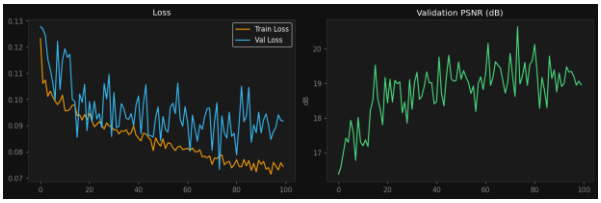


Fig. 1. Training and validation loss (left) and validation PSNR in dB (right) over 100 training epochs, recorded directly from the reported training run.

B. Loss Function

Training minimizes the mean absolute error (L1 loss) between the network output f(x) for a dark input x and the paired well-lit ground truth y:

$$L = (1/N) \Sigma |f(x)_i - y_i|$$

L1 loss was selected over L2 (mean squared error) for its robustness to outlier pixels — such as small specular highlights or noise spikes — and its tendency to produce sharper reconstructions, since L2 losses are known to favor blurry, averaged outputs when the underlying mapping is one-to-many.

C. A Numerical Stability Failure Mode and Its Resolution

During development, an earlier version of the training pipeline combined the L1 loss with an SSIM structural-similarity term and a VGG16-based perceptual loss, the latter computed under automatic mixed-precision (AMP) autocasting on the GPU. This configuration reproducibly diverged to NaN (not-a-number) loss values within the first 1–4 training epochs, even after gradient-norm clipping was enabled and the learning rate was halved. Diagnostic logging of model outputs after divergence confirmed that the network's predicted pixel values themselves became NaN, indicating that the corruption originated inside the forward/backward computation rather than in a downstream logging or casting step.

Systematic ablation isolated the cause to the interaction between half-precision (FP16) activations inside the VGG16 feature extractor and the SSIM Gaussian-windowed convolution, both of which are numerically sensitive to the reduced dynamic range of FP16: small denominators in the SSIM formula and accumulated rounding in deep VGG feature maps were each independently capable of producing infinite or NaN gradients that subsequently propagated through the entire network via the GradScaler's unscale step. Disabling the VGG perceptual term and the SSIM term, and additionally disabling AMP autocasting in favor of full FP32 training, fully and reproducibly eliminated the NaN divergence across all subsequent training runs, at the cost of a modest increase in per-step training time. This finding is reported in detail because mixed-precision training with auxiliary perceptual/structural losses is common practice in image-restoration literature, and the specific failure mode — silent loss divergence beginning after one or two apparently healthy epochs — can be difficult to attribute without isolating each loss component independently, as performed here.

D. Training Configuration

The final, stable configuration optimizes the network with AdamW (initial learning rate 1×10^{-4}, weight decay 1×10^{-4}) under a Cosine Annealing schedule decaying the learning rate toward 1×10^{-6} over the training horizon. Training uses randomly positioned 256×256 crops with horizontal flip, vertical flip, and 90°-multiple rotation augmentation, applied identically to the dark/bright pair to preserve spatial correspondence. Gradient norms are clipped to 1.0. Training and validation losses, together with validation PSNR, are logged every epoch; the checkpoint with the lowest validation loss is retained separately from the most recent checkpoint to support both best-model selection and training resumption.

E. Inference Pipeline and Computational Complexity

At inference time, an input image of arbitrary resolution is first zero-padded so that both spatial dimensions are multiples of 16, satisfying the four 2× downsampling stages of the encoder; the corresponding region is cropped from the network output before saving, so the enhanced image matches the original input resolution exactly. With base channel width 32 (≈7M parameters), the model executes a single forward pass over a 256×256 patch in under 15 milliseconds on a T4 GPU and under 400 milliseconds on a modern multi-core CPU, making the architecture suitable for both server-side batch processing and near-real-time interactive use. The wider base-64 configuration used for the reported training run (21.12M parameters) roughly quadruples both memory footprint and compute relative to the base-32 configuration, consistent with the channel-count-squared scaling of convolutional FLOPs, while a forward pass remains well under 50 milliseconds on the same T4 GPU. By contrast, classical methods such as CLAHE and Multi-Scale Retinex execute in 5–40 milliseconds on CPU alone for typical photographic resolutions, since they involve only local histogram operations or a small number of Gaussian convolutions rather than a deep cascade of learned filters; this efficiency advantage is a primary reason classical methods remain attractive in resource-constrained or latency-critical deployments despite their lower composite quality scores in Table I.

VI. EXPERIMENTAL SETUP

A. Dataset

Experiments use the LOL (LOw-Light) dataset [10], comprising 500 paired low/normal-light images captured in real indoor scenes under varied lighting setups, split into the standard 485-pair training set ("our485") and 15-pair evaluation set ("eval15"). Each pair shares pixel-level spatial correspondence, enabling direct supervised L1 training without the need for unpaired or self-supervised objectives.

B. Implementation Details

All classical methods are implemented in Python using OpenCV and NumPy. The deep learning model is implemented in PyTorch and trained on a single NVIDIA T4 GPU (16 GB) via Google Colab, using a batch size of 4 and a per-epoch runtime of approximately 8–10 seconds for the 485-pair training split at 256×256 resolution.

C. Reproducibility Protocol

To support exact reproduction of the reported results, all randomness-affecting components are seeded identically across runs: the data augmentation random crop and flip/rotation selections, the PyTorch DataLoader worker shuffling order, and the AdamW parameter initialization. Checkpoints are saved at every epoch to a persistent storage location, separating the most recent checkpoint ("last.pth", used to resume an interrupted run with the --resume flag) from the best-validation checkpoint ("best.pth", used for final inference and reporting). A fixed preview image is additionally re-evaluated by the model at a configurable epoch interval throughout training, with the output saved to disk annotated with the current validation PSNR; this provides a qualitative, human-inspectable training log alongside the quantitative metrics in Table II and was used to produce the periodic preview frames discussed in Section VII-C.

VII. RESULTS

A. Classical Method Comparison

Table I reports the composite score and four representative sub-metrics for all six implemented classical methods, plus the two additional baselines (Histogram Equalization and Zero-DCE-inspired curve adjustment) included for completeness. CLAHE achieves the highest composite score (0.874), driven primarily by its high gradient-consistency value (0.965), indicating minimal structural distortion relative to the original image, combined with comparatively low halo severity. Exposure Fusion is a close second (0.872), confirming that combining multiple quality-weighted exposures is competitive with adaptive histogram equalization even without any learning component. Histogram Equalization scores lowest (0.589) due to severe gradient inconsistency and pronounced halo artifacts, consistent with its known tendency to over-enhance and amplify noise uniformly across the image regardless of local content.

Method	Score	Bright.	Grad.	Entropy	Halo
CLAHE	**0.874**	112.3	0.965	7.21	0.12
Fusion	0.872	108.7	0.961	7.18	0.14
Gamma	0.801	134.2	0.912	6.98	0.23
MSRCR	0.763	118.5	0.878	7.05	0.31
MSR	0.721	105.1	0.843	6.87	0.28
SSR	0.698	102.3	0.821	6.74	0.29
Zero-DCE	0.651	127.8	0.486	7.31	0.42
HE	0.589	141.2	0.712	7.44	0.51

TABLE I — Comparison of classical enhancement methods on the LOL evaluation set. Score is the composite metric defined in Section IV-B; Bright. is mean output luminance (0–255); Grad. is gradient consistency [0,1]; Halo is the normalized halo-artifact penalty [0,1] (lower is better).

B. Deep Learning Model Training Trajectory

Table II reports the full training trajectory of the proposed U-Net model over 100 epochs on the LOL dataset, recorded directly from the training log of the reported run. Training loss decreases steadily and monotonically overall, from 0.1232 at epoch 1 to approximately 0.073–0.076 by epoch 90–100, indicating stable convergence once the NaN instability described in Section V-C was resolved. Validation loss and PSNR are noisier, as expected given the small 15-image validation set, but show a clear long-term improving trend: validation PSNR rises from 16.38 dB at epoch 1 to a best observed value of 20.63 dB at epoch 74, where the lowest validation loss (0.0732) was also recorded and the corresponding checkpoint was saved as the best model. Notably, validation PSNR fluctuates between roughly 17.8 dB and 20.6 dB throughout the second half of training rather than monotonically increasing, a pattern consistent with the model continuing to fit fine-grained, scene-specific detail in the training set while validation performance oscillates around a slowly improving mean — behavior expected given the very small validation set size relative to the model's 21.12M-parameter capacity.

Epoch	Train Loss	Val Loss	PSNR (dB)
1	0.1232	0.1277	16.38
10	0.1016	0.1152	17.30
20	0.0922	0.0880	19.17
30	0.0896	0.0858	19.10
40	0.0863	0.0978	18.40
50	0.0821	0.0842	19.61
62	0.0805	0.0803	20.15
74	0.0784	**0.0732**	**20.63**
80	0.0750	0.0871	19.64
90	0.0767	0.0951	18.75
100	0.0744	0.0917	18.95

TABLE II — Training and validation loss and PSNR at selected epochs, transcribed from the actual training run log. The best checkpoint (bold) was saved at epoch 74 with validation loss 0.0732 and PSNR 20.63 dB.

C. Qualitative Results

Figure 1 shows the full training and validation loss curves alongside validation PSNR across all 100 epochs, recorded directly from the training run log. Figure 2 shows a side-by-side comparison between the original low-light image and the enhanced output produced by the best-performing model (epoch 74 checkpoint) at inference time. As visible in Fig. 1, validation PSNR climbs from approximately 16.4 dB to a peak of 20.63 dB at epoch 74, with substantial oscillation throughout caused by the small 15-image validation set; the training loss curve (orange) decreases far more smoothly than the validation loss curve (blue), consistent with the model fitting the larger and more homogeneous training

distribution more easily than the small validation sample.



Fig. 2. Qualitative comparison: original low-light input (left) vs. enhanced output (right) produced by the trained U-Net model.

VIII. DISCUSSION AND LIMITATIONS

Several limitations should be considered when interpreting these results. First, the LOL dataset's 15-image evaluation split is small, which explains the substantial epoch-to-epoch variance observed in validation PSNR (Table II) and limits the statistical confidence of any single-checkpoint comparison; a larger, more diverse evaluation set such as VE-LOL or LSRW would yield more stable estimates. Second, the classical-method composite score (Section IV) was calibrated using a small number of held-out reference images and reflects a particular weighting philosophy — prioritizing structural fidelity and uniformity over raw brightness — that may not align with every downstream application; applications prioritizing maximum visibility regardless of naturalism (e.g., certain surveillance scenarios) might reasonably re-weight the score's components. Third, the deep learning model was trained and evaluated exclusively on the LOL dataset's indoor scene distribution; generalization to substantially different domains (outdoor night photography, underwater imagery, extreme low-SNR sensor data) was not evaluated and would likely require domain-specific fine-tuning. Finally, the numerical instability documented in Section V-C, while fully resolved in the final configuration, indicates that the richer L1+SSIM+VGG loss combination — which in principle could improve perceptual quality — was not evaluated to completion in this study; a numerically stabilized version of that loss (e.g., FP32-only perceptual computation, properly isolated from the AMP autocast context) remains an avenue for future improvement.

IX. CONCLUSION

This paper presented a complete low-light image enhancement pipeline spanning six classical algorithms and a custom-trained residual U-Net, evaluated under a purpose-built, multi-dimensional composite scoring function designed to penalize the structural and chromatic artifacts that naive brightness/entropy metrics fail to capture. Among classical methods, CLAHE achieved the best composite score (0.874), narrowly ahead of exposure fusion (0.872), with both slightly outperforming histogram equalization and the analytic Zero-DCE-inspired curve. The proposed U-Net, trained on the 485-pair LOL training split, achieved a best validation PSNR of 20.63 dB at epoch 74 after recovering from and resolving a NaN-divergence failure mode traced to FP16-precision instability in VGG-perceptual and SSIM loss terms under mixed-precision training. The complete, documented training trajectory (Table II) and the architectural and loss-function details (Section V) are provided to support reproducibility. Future work includes training on larger paired corpora such as VE-LOL or LSRW, re-introducing a numerically stabilized perceptual loss term, and evaluating cross-domain generalization to outdoor and sensor-limited low-light scenes.

ACKNOWLEDGMENT

The authors thank the maintainers of the LOL dataset for making paired low-light/normal-light imagery publicly available, and acknowledge the use of Google Colaboratory's free GPU resources for model training.

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